Business Statistics

Module 6
Video 6.1
Introduction to Sampling Theory

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Agenda

- Purpose
- Principles
- Method of Sampling
- Types of Sampling
- Sample Size
- Sampling and Non-Sampling Errors

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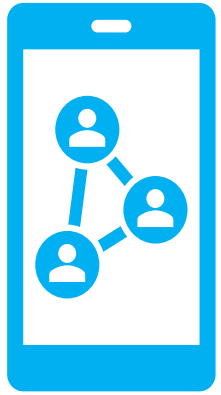
Why do we need Sampling?

Suppose you are conducting a study on the effect of social media use among teenagers



What is the percentage of teenagers that use social media?

What is the most commonly used social media platform?

Ideally you would want to collect data from **every** teenager to answer this accurately- but that is not possible!

This is where we use sampling!

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What is Sampling?

Sampling is a method of selecting a number of data from a larger defined population to represent the said population

https://www.freepik.com/free-vector/happy-citizens-forming-circle_1311976.htm#page=1&query=population&position=44

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Principles of Sampling

Principles of Sampling

- **Statistical Consistency**: When a large number of data is selected from a population, it is likely to have the same characteristics as the population
- **Large Sample Size***: The larger the number of sample data points selected from the population, the more accurate the estimation of population characteristics

* *Sample size: Number of data points included in the sample*

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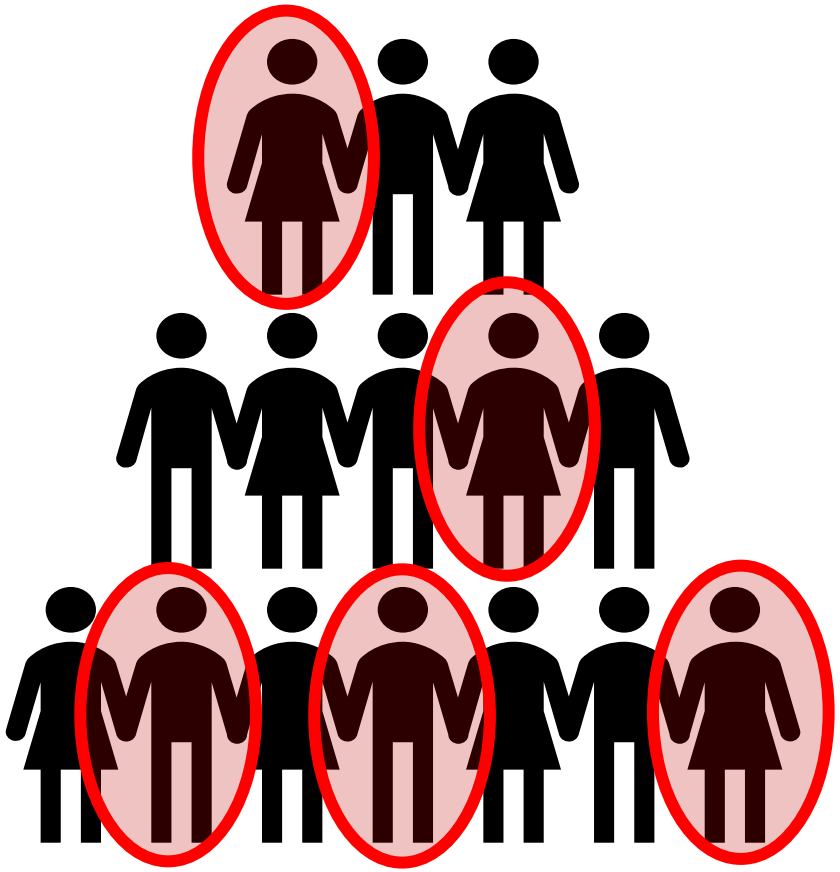
Types of Sampling

Simple Random Sampling



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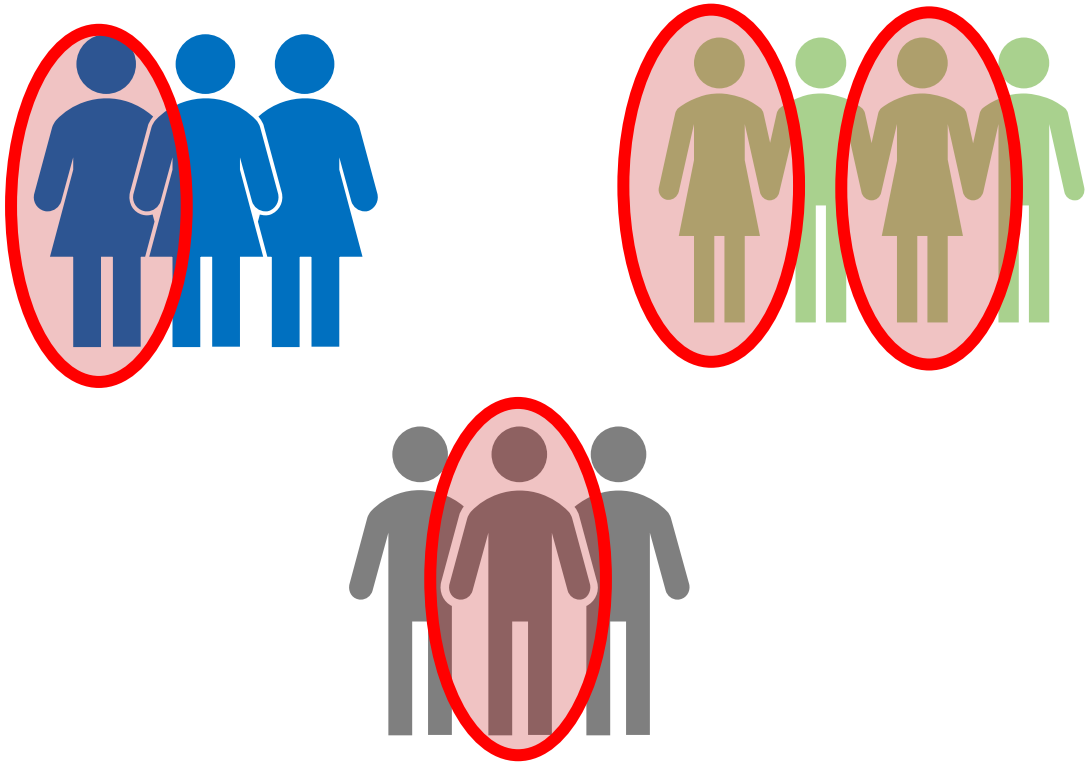
Types of Sampling

Stratified Random Sampling



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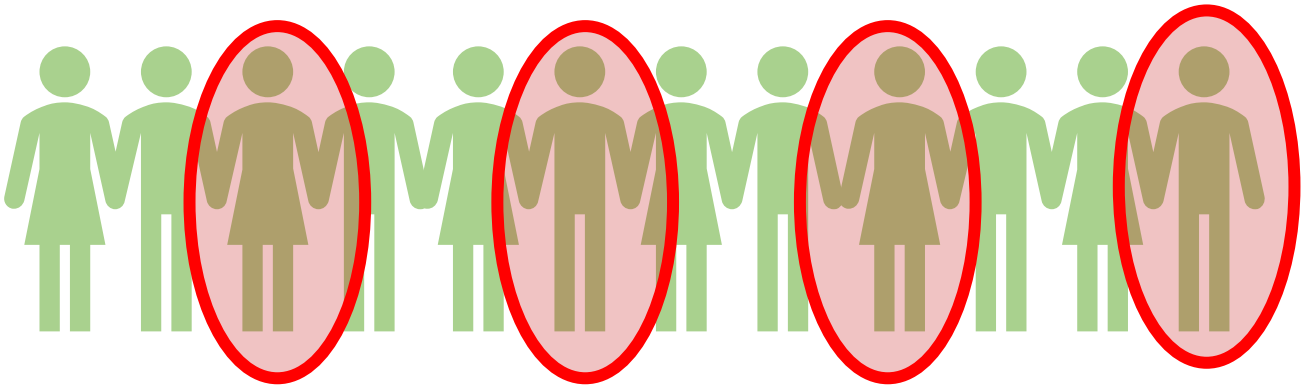
Types of Sampling

Systematic Sampling



E.g. Here we selected every 3rd element

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Sampling and Non-Sampling Errors

Sampling Errors

Error that arises from using the sample instead of the whole population

Non-Sampling Errors

Error that arises due to others factors in data collection

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Summary

- Definition of Sampling
- Principles of Sampling
- Types of Sampling
  - Simple Random
  - Stratified Random
  - Systematic
- Sampling and Non-Sampling Errors

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